1.0 Overview of the Project

Universiti Teknologi Malaysia, as one of the best technological universities in the world, has many students who move around every day from their residences to the various faculties. Therefore, efficient campus transportation is no longer just a convenience but a vital necessity. The university operates an internal shuttle bus service running different routes within the university campus connecting various residential colleges, faculties, and administration blocks.

However, the current bus service is not satisfactory to the students due to several reasons such as the lack of active buses with high student density. These issues lead to the inconvenience when using the bus service provided and may affect the daily study plan such as students being unable to attend the class on time due to the delay of the buses.

Thus, this project is to design and develop a new UTM Transport App to ensure that students can get transportation easily on campus. This mobile application focuses on bringing the physical and virtual sides of campus life together through the use of real-time data analysis and mobile applications to maximize efficiency and engagement within the campus ecosystem.

In this project, we will collect the students’ suggestions or even complaints, and analyse the pain points of the existing transport system. After that, the existing features in the current system will be modified and the new features will be created based on students' requirements. All of these features will be included in our UTM Transport App. Therefore, instead of switching various mobile applications, students can get satisfactory transportation by using only a single mobile application. We will also develop an admin portal for UTM Administrator to support our application to ensure we have a complete UTM Transport System.

2.0 Problem Statement

At a campus as huge as Universiti Teknologi Malaysia (UTM), transportation is not just a "nice-to-have" feature, it is the heartbeat of our daily academic lives. However, the current transportation system tends to be fragmented and traditional that just cannot keep up with the intense schedule of a modern student. For most of us, arranging our traveling schedule is not the smooth transition it should be. It has become a daily source of stress and friction that drains our focus and productivity. To alleviate this issue, we need to address five specific "pain points" that are currently making life harder for the UTM community:

1. **The Bus Arrival Uncertainty**

   Right now, the biggest headache is the total lack of transparency. We are forced to stand at bus stops for who-knows-how-long, often sweating under the scorching sun or getting soaked in a heavy rain, without any idea if the bus is two minutes away or twenty. This uncertainty causes a domino effect, such that we end up late for crucial lectures, miss the start of lab sessions, and live in a constant state of "transit anxiety" that makes it impossible to plan our study groups or even a simple lunch break effectively.

2. **The Peak-Hour Seat Shortages**

   During the morning and evening rush, the campus transport system basically turns into "survival of the fittest" mode. This is because we have no way to check how full a bus is in advance, we often have to watch bus after bus drive past because they are packed with people from the previous stations. It is not just about the delay, but also about the discomfort of being squeezed into overcrowded vehicles and the safety risks that come with it, leaving those at later stops completely stranded.

3. The Inefficient Carpooling

So many of us are willing to share rides or land a helping hand to ease the load on the buses, but there is not a safe, official way to make that happen. We are currently stuck using disorganized social media chats or word-of-mouth, which is awkward and unreliable. This "fragmented carpooling" means thousands of seats go empty every day while other students are left waiting at the station or walking with their bare feet, wasting a huge opportunity to build a better campus community and lower our carbon footprint.

4. No Route Planning Tools

UTM is huge, and for "freshies" or visitors, it can feel like a total maze. Without a smart tool to help plan a route, students struggle to find the fastest way to get from their residential colleges to distant faculties or admin buildings. We end up wasting time taking long, inefficient paths or walking much further than we need to, simply because we do not have a map that shows us how to navigate the transit transfers correctly.

5. The Lack of Feedback

The most frustrating part of all this is that when things go wrong, we have no real voice. There is not a dedicated, systematic way to report things like a bus being consistently late, a vehicle being dirty, or safety concerns with driving. Without this feedback, the university admin is essentially "flying blind," making it impossible for them to see where the service is failing or how to fix it. This silence leads to a cycle of the same problems happening semester after semester with no end in sight.

3.0 Proposed Solutions

- **Design a Secure User Management Module**
  This module focuses on account management functions such as account creation, authentication, and user-profile maintenance to ensure that users of the application are indeed verified UTM students. They need to sign up using UTM email. So, users just need to register their account and click on the "Remember Me" checkbox if one wants quick future logins.

- **Implement a Live Tracking & Mapping Module**
  This module offers users the ability to view live information on campus buses' routes and locations. The information displayed consists of the actual location of the bus on the map, along with an Estimated Time of Arrival (ETA) at their current station. The module enables users to pick any bus on the map and observe its movement, along with its plate number.

- **Implement a Seat Availability Monitor Module**
  This module allows users to view real-time seat availability of approaching campus buses. The information displayed includes the current occupancy level (available, moderate, or full). So, users can select a bus and check its seat status along with its route, helping them decide whether to board or wait for the next bus.

- **Create a Carpool Module**
  This module helps in making travelling on the campus flexible since it offers the students the ability to form rides with other students. Users have two options in this module either to form their own carpool ride or join others. Users can go through the list of available rides and find someone who matches.

- **Design a Route Planning Module**

  This module acts as the smart guide in planning how users can travel on the campus in the best way. The fastest route options will be provided together with step by step instructions as well as any required bus transfer along the way. Users can simply input the destination point from the point of departure to access various route options.

- **Implement a Feedback & Rating Module**

  This module allows the users to help in improving campus transport through feedback. Here, users can rate their experience riding the bus and report problems such as lateness and overcrowding among others. Once the trip ends, a rating window pops out where users can rate the bus.

4.0 Current Business Process / Workflow

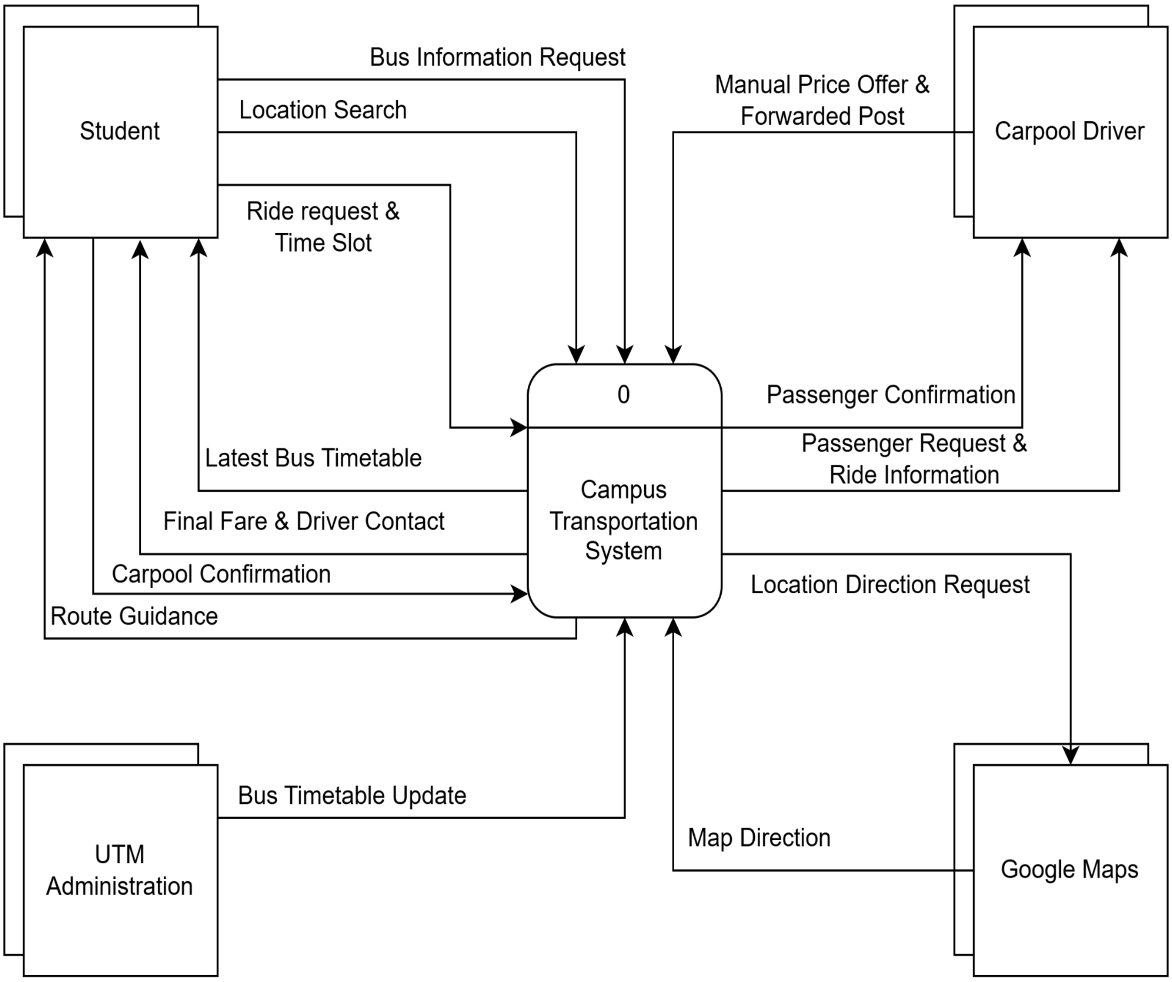
Currently, navigating the huge Universiti Teknologi Malaysia (UTM) campus relies heavily on traditional, manual coordination and fixed timetable shuttle buses. When a student needs to travel from their residential college to a faculty building, they walk to the nearest designated bus stop and wait. Due to there is no live tracking visibility into the transit network, the student must rely on guesswork, often standing under the scorching hot sun or getting soaked during sudden heavy rains, unsure if the bus is two minutes or twenty minutes away.

During peak morning and evening hours, the situation deteriorates. When a bus finally arrives, the student frequently discovers it is already packed with passengers from the previous stops. Unable to board, the student is forced to wait for the next uncertain bus, resulting in a domino effect of missed lectures, late lab sessions, and high “transit anxiety”. If a student decides to seek alternative transport, they must resort to fragmented, unofficial carpooling. They open disorganized social media groups or messaging apps (like Telegram) to manually ask for rides, which is inefficient and poses security concerns as driver identities are not formally verified.

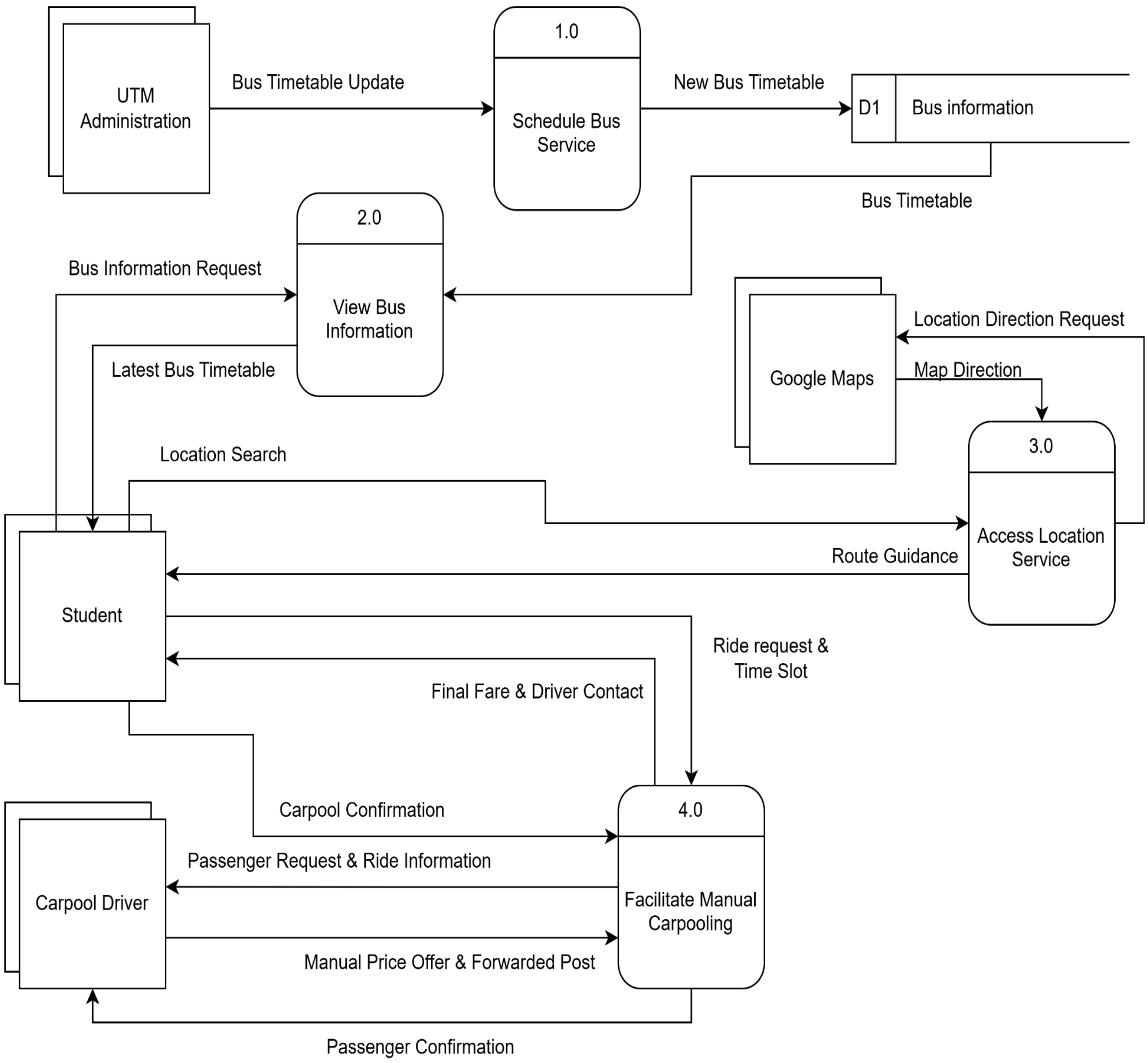
Furthermore, for freshmen or visitors navigating the campus maze, there is no centralized tool to plan their journey. They often take long, inefficient walking paths or miss necessary bus transfers. Finally, when the journey concludes, if the students experience a delayed schedule, reckless driving, or a dirty vehicle, there is no dedicated channel to report it. The administration remains unaware of these daily operational failures, and the cycle of frustration continues into the next semester.

5.0 Logical DFD AS-IS System

Context Diagram



<u>Diagram 0</u>



UTM Administration
Bus Timetable Update
1.0
Schedule Bus Service
New Bus Timetable
D1
Bus information
Bus Timetable
2.0
View Bus Information
Bus Information Request
Latest Bus Timetable
Google Maps
Location Direction Request
Map Direction
3.0
Access Location Service
Location Search
Route Guidance
Student
Ride request & Time Slot
Final Fare & Driver Contact
Carpool Confirmation
4.0
Facilitate Manual Carpooling
Carpool Driver
Passenger Request & Ride Information
Manual Price Offer & Forwarded Post
Passenger Confirmation

<u>Child Diagram</u>